[MP2] Testcase 7-10 Race condition 問題？

2 2

謝典樺 (HSIEH, TIEN-HUA)

All Sections

我不太確定是Race condition的問題還是我自己的實現有誤。當我在跑Testcase 7到10的時候，有時候會如以下顯示FAIL。

```
== Test public/mp2-7 (3%) ==
public/mp2-7: FAIL (0/3) (1.1s)
    Invalid max_objs: set to 8 bus is actually 6
    QEMU output saved to out/public/mp2-7.out.failed
== Test public/mp2-8 (3%) ==
public/mp2-8: FAIL (0/3) (1.2s)
    Invalid max_objs: set to 8 bus is actually 6
    QEMU output saved to out/public/mp2-8.out.failed
== Test public/mp2-9 (3%) ==
public/mp2-9: FAIL (0/3) (1.2s)
    Invalid max_objs: set to 8 bus is actually 7
    QEMU output saved to out/public/mp2-9.out.failed
== Test public/mp2-10 (3%) ==
public/mp2-10: FAIL (0/3) (1.5s)
    Invalid max_objs: set to 8 bus is actually 7
    QEMU output saved to out/public/mp2-10.out.failed
```

但有時候又可以PASS，看了輸出失敗的時候都是fileclose的訊息打亂了輸出，而成功的時候就沒有這個問題。

```
[SLAB] kmem_cache { name: file, object_size: 504, at: 0x0000000087f59000, in_cache_obj: 0 }
[SLAB]  [ full  slabs ]
[SLAB]      [ slab 0x0000000087f4e000 ] { freelist: 0x0000000000000000, inuse: 8, nxt: 0x0000000000000000, prev: 0x0000000000000000 }
[FILE] fileclose
[SLAB]          [ idx 0 ] { addr: 0x0000000087f4e028, as_ptr: 0x0000000f00000003, as_obj: {tp: 3, ref: 15, readable: 1, writable: 1, pipe: 0x0000000000000000, ip: 0x0000000080035488, off: 0, major: 1} }
[SLAB]          [ idx 1 ] { addr: 0x0000000087f4e220, as_ptr: 0x0000000000000000, as_obj: {tp: 0, ref: 0, readable: 1, writable: 1, pipe: 0x0000000000000000, ip: 0x0000000080035488, off: 0, major: 1} }
[SLAB]          [ idx 2 ] { addr: 0x0000000087f4e418, as_ptr: 0x0000000100000001, as_obj: {[FILE] fileclose
tp: 1, ref: 1, readable: 1, writable: 0, pipe: 0x0000000087f24000, ip: 0x0000000000000000, off: 0, major: 0} }
[SLAB]          [ idx 3 ] { addr: 0x0000000087f4e610, as_ptr: 0x0000000000000000, as_obj: {tp: 0, ref: 0, readable: 0, writable: 1, pipe: 0x0000000087f24000, ip: 0x0000000000000000, off: 0, major: 0} }
[SLAB]          [ idx 4 ] { addr: 0x0000000087f4e808, as_ptr: 0x0000000100000001, as_obj: {tp: 1, ref: 1, readable: 1, writable: 0, pipe: 0x0000000087eeb000, ip: 0x0000000000000000, off: 0, major: 0} }
[SLAB]          [ idx 5 ] { addr: 0x0000000087f4ea00, as_ptr: 0x0000000000000000, as_obj: {tp: 0, ref: 0, readable: 0, writable: 1, pipe: 0x0000000087eeb000, ip: 0x0000000000000000, off: 0, major: 0[FILE] fileclose
} }
[SLAB]          [ idx 6 ] { addr: 0x0000000087f4ebf8, as_ptr: 0x0000000100000001, as_obj: {tp: 1, ref: 1, readable: 1, writable: 0, pipe: 0x0000000087eab000, ip: 0x0000000000000000, off: 0, major: 0} }
[SLAB]          [ idx 7 ] { addr: 0x0000000087f4edf0, as_ptr: 0x0000000000000000, as_obj: {tp: 0, ref: 0, readable: 0, writable: 1, pipe: 0x0000000087eab000, ip: 0x0000000000000000, off: 0, major: 0} }
[SLAB]  [ partial   slabs ]
[SLAB]      [ slab 0x0000000087ecd000 ] { freelist: 0x0000000087ecd610, inuse: 3, nxt: 0x0000000000000000, prev: 0x0000000000000000 }
[SLAB]          [ idx 0 ] { addr: 0x0000000087ecd028, as_ptr: 0x0000000100000002, as_obj: {tp: 2, ref: 1, readable: 0, writable: 1, pipe: 0x0000000000000000, ip: 0x0000000080035510, off: 0, major: 0} }
[SLAB]          [ idx 1 ] { addr: 0x0000000087ecd220, as_ptr: 0x0000000000000000, as_obj: {tp: 0, ref: 0, readable: 1, writable: 1, pipe: 0x0000000000000000, ip: 0x0000000080035598, off: 0, major: 0} }
[SLAB]          [ idx 2 ] { addr: 0x0000000087ecd418, as_ptr: 0x0000000000000000, as_obj: {tp: 0, ref: 0, readable: 1, writable: 1, pipe: 0x0000000000000000, ip: 0x0000000080035620, off: 0, major: 0} }
[SLAB]          [ idx 3 ] { addr: 0x0000000087ecd610, as_ptr: 0x0000000087ecd808, as_obj: {tp: -2014521336, ref: 0, readable: 5, writable: 5, pipe: 0x0505050505050505, ip: 0x0505050505050505, off: 84215045, major: 1285} }
[SLAB]          [ idx 4 ] { addr: 0x0000000087ecd808, as_ptr: 0x0000000087ecda00, as_obj: {tp: -2014520832, ref: 0, readable: 5, writable: 5, pipe: 0x0505050505050505, ip: 0x0505050505050505, off: 84215045, major: 1285} }
[SLAB]          [ idx 5 ] { addr: 0x0000000087ecda00, as_ptr: 0x0000000087ecdbf8, as_obj: {tp: -2014520328, ref: 0, readable: 5, writable: 5, pipe: 0x0505050505050505, ip: 0x0505050505050505, off: 84215045, major: 1285} }
[SLAB]          [ idx 6 ] { addr: 0x0000000087ecdbf8, as_ptr: 0x0000000087ecddf0, as_obj: {tp: -2014519824, ref: 0, readable: 5, writable: 5, pipe: 0x0505050505050505, ip: 0x0505050505050505, off: 84215045, major: 1285} }
[SLAB]          [ idx 7 ] { addr: 0x0000000087ecddf0, as_ptr: 0x0000000000000000, as_obj: {tp: 0, ref: 0, readable: 5, writable: 5, pipe: 0x0505050505050505, ip: 0x0505050505050505, off: 84215045, major: 1285} }
[SLAB] print_kmem_cache end
[FILE] fileclose
[FILE] fileclose
[FILE] fileclose
[FILE] fileclose
$ qemu-system-riscv64: terminating on signal 15 from pid 57876 (make)
```

我想請問這是什麼問題呢？

Search entries or author | Unread | Subscribe

Reply



李尚哲 (LI,SHANG-JHE) (https://cool.ntu.edu.tw/courses/46296/users/188748)

Mar 26, 2025

一個是FILE盡量不要輸出，一個是print kmem cache要鎖起來，雖然我18還是有時候不會過。

Reply



楊子慶 (YANG, TZU-CHING) (https://cool.ntu.edu.tw/courses/46296/users/229158)

Mar 26, 2025

同學您好

助教的直譯器保證同學在 `print_kmem_cache` 、 `kmem_cache_create` 、 `kmem_cache_alloc` 、 `kmem_cache_free` 時輸出 `[FILE]` 或者其他訊息非 `[SLAB]` 開頭的訊息不會影響正確性檢查，故並非該問題。

根據同學的錯誤提示：

```
Invalid max_objs: set to 8 but (抱歉 bus 是助教打錯字) is actually <not 8>
```

可知同學在列印 Spec 所規範的資訊中並未將 slab 中所有物件的資訊印出，請同學將所有資訊印出。

希望以上回覆有解答到同學的疑問。
若有更多問題請不吝發問，或者於今天或週五的助教時間前來討論。

子慶

Reply